

Solution Requirements

Date	29 June 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S. No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	User Input	<i>Allow users to enter an event description and networking interests.</i>	High
2	Theme Extraction	<i>Extract key themes from the event description using the DistilBERT model.</i>	High
3	Conversation Starter Generation	<i>Generate personalized conversation starters using GPT-2 based on extracted themes and user interests.</i>	High

4	Fact Verification	<i>Verify user queries by retrieving reliable summaries from the Wikipedia API.</i>	High
5	Interaction History	<i>Save generated conversation starters and feedback history locally for future reference.</i>	Medium
6	User Interface	<i>Provide an interactive and responsive Streamlit interface for user interaction.</i>	High
7	Backend API	<i>Process requests using FastAPI and return responses efficiently.</i>	High
8	Error Handling	<i>Display meaningful error messages for invalid inputs or API failures.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S. No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance	<i>Generate conversation starters quickly after user input.</i>	<i>Response time less than 5 seconds (excluding first model loading).</i>
2	Scalability	<i>Support multiple user requests without affecting application performance.</i>	<i>Stable performance for concurrent users.</i>
3	Security	<i>Protect environment variables and validate user inputs.</i>	<i>Sensitive data stored in .env; input validation implemented</i>
4	Reliability	<i>Ensure consistent generation of conversation starters and fact verification.</i>	<i>Successful execution rate greater than 95%.</i>
5	Usability	<i>Provide a simple, intuitive, and responsive user interface.</i>	<i>Easy navigation with minimal user training.</i>
6	Maintainability	<i>Follow modular architecture with separate frontend, backend, AI services, and utilities.</i>	<i>Code organized into reusable modules with documentation.</i>